# CariSurg MedTech Pathways Research Project

# Week 5 - Feasibility Memo


**Project Title:** Enhancing Emergency Department Triage with Artificial Intelligence within the Caribbean Context

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**Student GitHub:** https://github.com/imaniiz/CariSurg-Portfolio



**Executive Summary:** This memo evaluates the suitability of the Mercer emergency department dataset for developing a baseline AI-assisted triage model. The assessment focuses on data quality, completeness, clinical relevance and limitations identified during exploratory data analysis.


## Feasibility Memo:

### Section 1: Verdict:

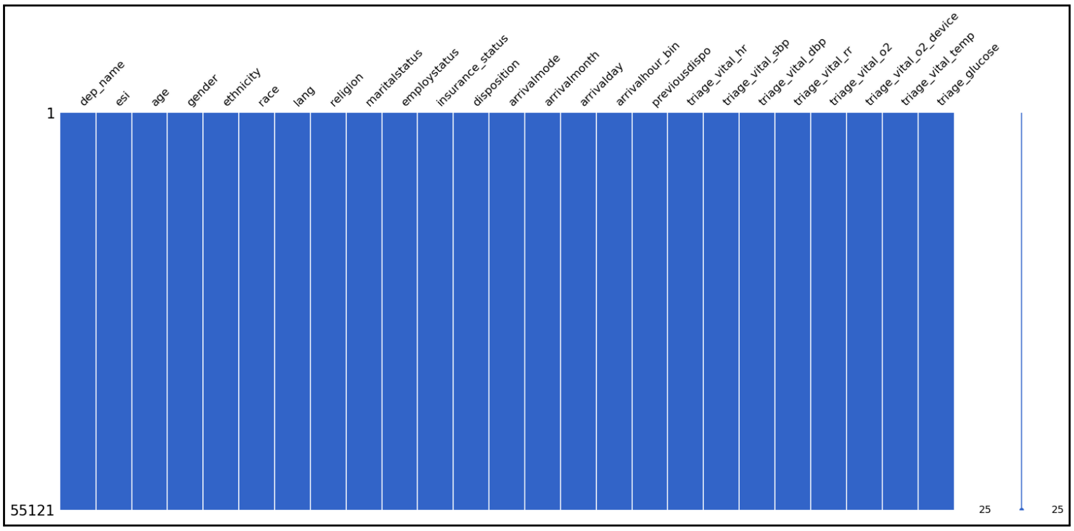
The dataset is suitable for developing a baseline AI-assisted triage model, provided that leakage variables are excluded and the limitations relating to generalisability are acknowledged

### Section 2: Dataset Summary

The dataset comprises 55,121 emergency department patient encounters and contains 225 clinical features collected during the patient's visit to the emergency department. These variables include demographic information, administrative details, triage vital signs and approximately 200 binary chief complaint flags indicating whether a patient presented with a particular symptom.

The target variable for this project is the Emergency Severity Index (ESI), a five-level triage system used to classify patients according to the urgency of their condition and the anticipated resources required for treatment. The patient population consists of adults aged 18 years and older, with demographic variables including age, gender, race, ethnicity, language, marital status, employment status and insurance status. Clinical information recorded at triage includes heart rate, respiratory rate, systolic and diastolic blood pressure, oxygen saturation, body temperature and blood glucose level, providing a comprehensive representation of each patient's physiological condition upon arrival. Initial exploratory data analysis indicates that the structured portion of the dataset is highly complete, with no missing values identified in the structured demographic with no missing values identified in the structured demographic, administrative and vital sign variables. Furthermore, all measured vital signs fall within clinically plausible ranges, although some statistical outliers are present, reflecting the natural variability expected within an emergency department population rather than obvious data entry errors. The distribution of ESI classes is also consistent with typical emergency department activity, with the majority of patients assigned to ESI Levels 2 and 3.

Figure 1: Missingness across Structured Variables




## Section 3: Top 3 Data Quality Concerns

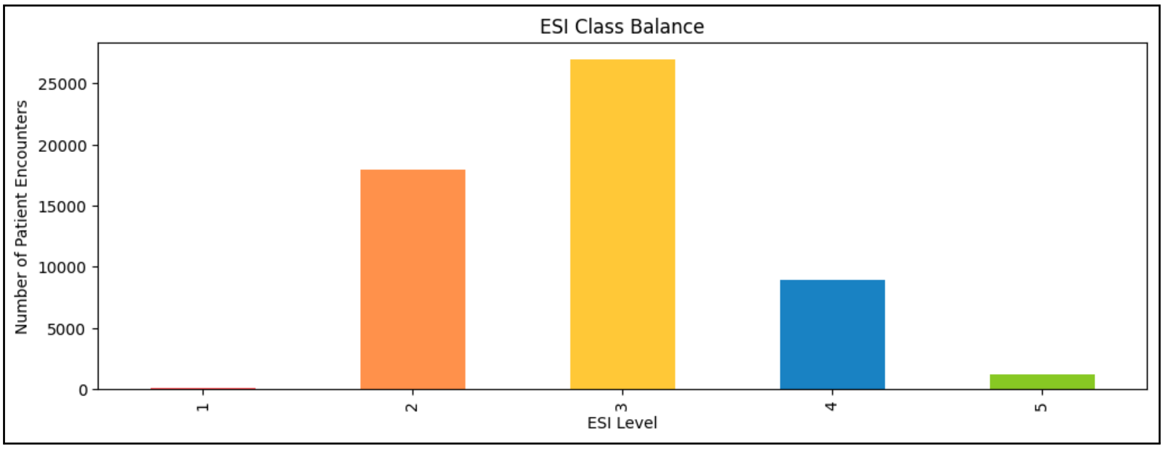
### Concern 1: Outcome Leakage

One limitation identified during the exploratory analysis is the presence of variables that would not be available at the time of triage. Variables such as *disposition* and *previous disposition* describe the patient's outcome after clinical assessment and treatment has occurred. Including these variables during model training would introduce outcome leakage, allowing the model to learn information that would not be available when making real-time triage decisions. To ensure that the model reflects the information available to triage clinicians, these variables should be excluded before model development. Removing leakage variables will produce a more realistic estimate of model performance and improve the likelihood that the model can generalise to real clinical practice.

### Concern 2: Class Imbalance

The distribution of Emergency Severity Index (ESI) levels is uneven across the dataset. ESI Level 2 and 3 dominate the dataset with approximately 49% and 32.5% encounters respectively, while the other ESI levels make up the remaining 18.5%. Although this distribution is typical of an emergency department, this imbalance may bias a machine learning model towards the more common triage categories, reducing its ability to accurately identify critically ill patients. Appropriate evaluation metrics and techniques such as class weighting should therefore be considered during model development in Week 6.

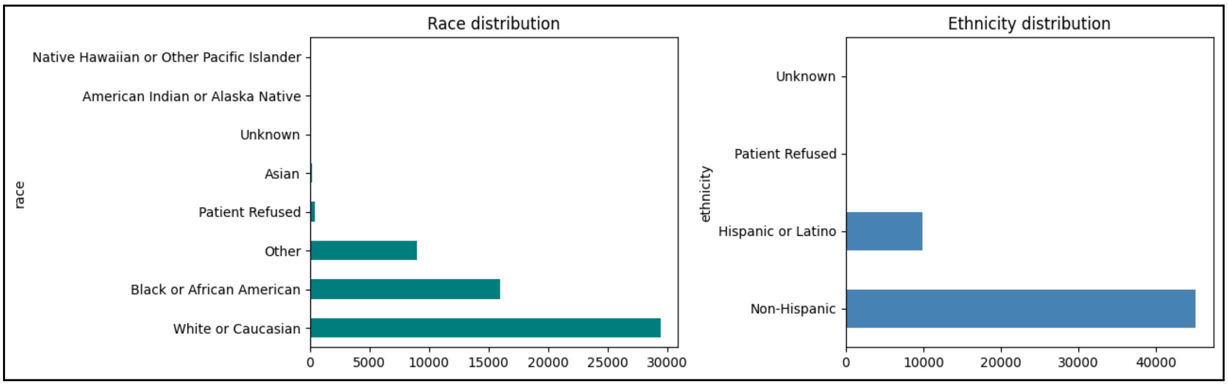
Figure 2: Bar Graph Showing ESI Class Balance




**Concern 3: Limited Generalisability**

Although the dataset is large and has significant variation, it originates from a single United States based emergency department. Patient demographics, disease prevalence, healthcare workflows and resource availability may differ from those encountered in Caribbean emergency departments. Any model developed using this dataset should be considered a baseline proof of concept rather than a model ready for deployment within Caribbean healthcare settings, External validation using local clinical data will be necessary before implementation.

Figure 3: Demographic Distribution




## Section 4: Top 3 Reasons to Process

**Reason 1: Excellent Data Completeness**

The structured demographic, administrative and triage variables demonstrate excellent completeness, with no missing values identified during exploratory analysis. This minimises the need for data imputation and provides a reliable foundation for model development. The high-quality structured data reduces the likelihood of introducing bias during preprocessing and simplifies the initial modelling process.

**Reason 2: Rich Clinical Information**

The dataset contains 225 patient variables, including demographic information, administrative details, seven routinely collected triage vital signs and approximately 200 binary chief compliant indicators. Together, these variables capture both the patient's physiological condition and presenting complaint, providing a clinically meaningful representation of emergency department presentations. The distributions of the recorded vital signs are consistent with expected emergency department populations and no clinically impossible measurements were identified after applying predefined physiological limits.

- Availability of real ESI labels
- Rich collection of demographic and clinical variables
- Comprehensive triage vital signs
- Large number of patient encounters
- Well-documented and reproducible data cleaning process
- Supporting figures such as vital sign distributions and chief complaint frequencies

**Reason 3: Large Sample Size**

With 55,121 patient encounters, the dataset provides a substantial number of observations for exploratory analysis and baseline machine learning. The large sample size improves the likelihood that common emergency presentations are adequately represented and supported and supports more stable model training compared with smaller datasets.

**<u>Section 5: Caveats</u>**

Although the dataset is suitable for baseline model development, several limitations should be acknowledged.

1. Temperature measurements are recorded in Fahrenheit, requiring unit awareness during preprocessing and interpretation
2. The chief complaint variables are highly sparse, with approximately 200 binary indicators where most values are zero for any individual patient
3. Feature selection techniques will be required to identify the most informative complaints before modelling

In addition, the correlations identified during exploratory analysis describe statistical associations rather than causal clinical relationships. Features that appear strongly associated with ESI should therefore be interpreted as potential predictors rather than direct determinants of triage severity. Finally, because the dataset originates from a single United States based emergency department, further validation using Caribbean emergency department data will be required before considering clinical deployment.

**Section 6: Top 10 Candidate Predictive Features**

The exploratory analysis identified several variables that are likely to contribute to predicting Emergency Severity Index (ESI) levels. The shortlist below combines clinical reasoning with the correlation analysis performed during data exploration. While no single feature demonstrates a strong linear relationship with ESI, the selected variables represent clinically meaningful indicators of patient acuity and are expected to provide greater predictive value when used together in a multivariable machine learning model.

Table 1: Candidate Predictive Features

| Rank | Feature | Correlation to ESI | Justification |
|---|---|---|---|
| 1 | Oxygen Saturation | 0.18 | Demonstrated the strongest association with ESI among the recorded vital signs and key indicators of respiratory compromise. |
| 2 | Chest Pain | -0.16 | One of the strongest chief compliant associations with ESI and clinically associated with life-threatening conditions |
| 3 | Shortness of Breath | -0.15 | Strong statistical association with ESI and an important indicator of respiratory distress |
| 4 | Respiratory Rate | -0.10 | Abnormal respiratory rate is an early physiological marker of deterioration |
| 5 | Heart Rate | -0.10 | Elevated heart rate reflects physiological stress, infection or shock and contributes to overall patient acuity |
| 6 | Altered Mental State | -0.13 | Although less common, altered mental status is clinically recognised as a high-priority presentation |
| 7 | Suicidal Ideation | -0.14 | Frequently receives urgent psychiatric assessment |
| 8 | Blood Pressure (SBP/DBP) | Weak individual correlations | Blood pressure showed limited correlation with ESI but remains a core triage vital sign |
| 9 | Age | Clinical importance | Although not included in the correlation matrix, age is widely recognised as an important predictor of patient risk |
| 10 | Arrival Mode | Clinical importance | Patients arriving by ambulance are generally more likely to require urgent assessment and higher-acuity care than walk-in patients |